Algebra Polynomials & Factoring Cheat Sheet

Unit Focus: *classifying, operating with, and factoring polynomials*

1 KEY VOCABULARY

monomial	A polynomial with one term.	$5x^3$
binomial	A polynomial with two terms.	$2x - 7$
trinomial	A polynomial with three terms.	$x^2 + x - 6$
polynomial	A sum or difference of one or more terms, where each term has a whole number exponent on the variable.	$4x^3 - 2x + 9$
degree	The highest exponent of the variable in a polynomial (in standard form).	Degree of $3x^4 - 2x + 1$ is 4.
leading coefficient	The numerical coefficient of the term with the highest degree.	In $-6x^5 + 2x^2 - 3$, the leading coefficient is -6.

2 CLASSIFYING POLYNOMIALS

By Number of Terms

Monomial	1 term	$7x^2$
Binomial	2 terms	$x - 4$
Trinomial	3 terms	$x^2 + 3x + 1$
Polynomial (more than 3)	4 or more terms	$2x^4 - x^3 + x - 5$

By Degree (Highest Exponent)

Constant (degree 0)	5
Linear (degree 1)	$3x - 2$
Quadratic (degree 2)	$4x^2 + x + 1$
Cubic (degree 3)	$x^3 - 2x$
Quartic (degree 4)	$2x^4 + 3x^2 - 7$

A polynomial can be classified by the number of terms and by its degree!

3 ADDING AND SUBTRACTING POLYNOMIALS

Combine like terms to add or subtract polynomials.

Example: $(3x^2 + 2x - 5) + (x^2 - 4x + 7)$

Step 1: Remove parentheses. $3x^2 + 2x - 5 + x^2 - 4x + 7$

Step 2: Combine like terms. $(3x^2 + x^2) + (2x - 4x) + (-5 + 7)$

Step 3: Simplify. $4x^2 - 2x + 2$

Answer: $4x^2 - 2x + 2$

4 MULTIPLYING A MONOMIAL BY A POLYNOMIAL

Distribute the monomial to each term in the polynomial.

Example: $3x(2x^2 - x + 4)$

Step 1: Distribute $3x$ to each term.

$3x(2x^2) + 3x(-x) + 3x(4)$

Step 2: Multiply. $6x^3 - 3x^2 + 12x$

Answer: $6x^3 - 3x^2 + 12x$

5 MULTIPLYING BINOMIALS

Use the distributive property (FOIL) or the area model.

Example: $(x + 3)(x + 5)$

Method 1: Distributive Property (FOIL)

$(x + 3)(x + 5)$
$= x(x) + x(5) + 3(x) + 3(5)$
$= x^2 + 5x + 3x + 15$
$= x^2 + 8x + 15$

Method 2: Area Model

	x	$+5$
x	x^2	$5x$
$+3$	$3x$	15

Answer: $x^2 + 8x + 15$

6 SPECIAL PRODUCTS

Use these patterns to multiply certain binomials quickly.

Square of a Sum

$(a + b)^2 = a^2 + 2ab + b^2$

Square of a Difference

$(a - b)^2 = a^2 - 2ab + b^2$

Example: $(x + 4)^2$

Step 1: Identify a and b. $a = x,\ b = 4$

Step 2: Apply the formula. $(x + 4)^2 = x^2 + 2(x)(4) + 4^2$

Step 3: Simplify. $x^2 + 8x + 16$

Answer: $x^2 + 8x + 16$

7 QUICK REMINDERS & COMMON MISTAKES

DO REMEMBER:

- Combine like terms only (same variable and same exponent).
- Follow the distributive property carefully.
- Keep track of signs, especially when subtracting.
- Write polynomials in standard form (descending powers).
- Use special product formulas when possible.
- Check your work by simplifying.

COMMON MISTAKES:

- Combining unlike terms (e.g., $3x + 2x^2$).
- Forgetting to distribute to every term.
- Not changing signs correctly when subtracting.
- Writing terms out of order (not in descending degree).
- Misapplying special product formulas.
- Arithmetic errors (e.g., $2x + 3x = 5x$, not $6x$).

Write final answers in standard form with no like terms left to combine!

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More Practice: *factoring strategies and finding zeros*

1 GCF FACTORING

Find the greatest common factor (GCF) of all terms and factor it out.

Example: Factor $6x^3 + 9x^2$.

Step 1: Find the GCF: $3x^2$ (GCF of $6x^3$ and $9x^2$).

Step 2: Divide each term by $3x^2$.

$$6x^3 \div 3x^2 = 2x$$
$$9x^2 \div 3x^2 = 3$$

Step 3: Factor out the GCF.

$$6x^3 + 9x^2 = 3x^2(2x + 3)$$

Answer: $3x^2(2x + 3)$

2 FACTORING TRINOMIALS $x^2 + bx + c$

Find two numbers that multiply to c and add to b.

Example: Factor $x^2 + 7x + 12$.

Step 1: Find two numbers that multiply to 12 and add to 7. The numbers are 3 and 4.

Step 2: Write the factors.

$$x^2 + 7x + 12 = (x + 3)(x + 4)$$

Look for two numbers with the correct product and sum!

Answer: $(x + 3)(x + 4)$

3 FACTORING $ax^2 + bx + c$

Use the AC method or trial and error to find two numbers that multiply to ac and add to b.

Example: Factor $2x^2 + 7x + 3$.

Step 1: Find ac: $2 \cdot 3 = 6$.

Step 2: Find two numbers that multiply to 6 and add to 7. The numbers are 6 and 1.

Step 3: Rewrite and factor by grouping.

$$\begin{aligned} 2x^2 + 7x + 3 &= 2x^2 + 6x + x + 3 \\ &= 2x(x + 3) + 1(x + 3) \\ &= (2x + 1)(x + 3) \end{aligned}$$

Answer: $(2x + 1)(x + 3)$

4 DIFFERENCE OF SQUARES

A difference of squares factors as:

$$a^2 - b^2 = (a - b)(a + b)$$

Example: Factor $x^2 - 25$.

Step 1: Recognize a difference of squares:

$$x^2 - 25 = x^2 - 5^2.$$

Step 2: Use the formula.

$$x^2 - 25 = (x - 5)(x + 5)$$

Both terms must be perfect squares and there is a minus sign.

Answer: $(x - 5)(x + 5)$

5 FACTORING BY GROUPING

Group terms, factor out the GCF from each group, then factor out the common binomial.

Example: Factor $x^3 + 3x^2 + 2x + 6$.

Step 1: Group the terms.

$$(x^3 + 3x^2) + (2x + 6)$$

Step 2: Factor the GCF from each group.

$$x^2(x + 3) + 2(x + 3)$$

Step 3: Factor out the common binomial.

$$(x + 3)(x^2 + 2)$$

Look for a common binomial after factoring each group.

Answer: $(x + 3)(x^2 + 2)$

6 SOLVING QUADRATICS BY FACTORING

Factor the quadratic, set each factor equal to zero, and solve for the zeros (solutions).

Example: Solve $x^2 + 7x + 12 = 0$.

Step 1: Factor the quadratic.

$$x^2 + 7x + 12 = (x + 3)(x + 4)$$

Step 2: Set each factor equal to zero.

$$x + 3 = 0 \quad | \quad x + 4 = 0$$
$$x = -3 \quad | \quad x = -4$$

The zeros are the x-values that make the equation equal to zero.

Answer: $x = -3$ or $x = -4$

7 FACTOR CHECK

Multiply the factors to make sure you get the original expression.

Example: Check the factorization $(x + 3)(x + 4) = x^2 + 7x + 12$.

Step 1: Multiply using FOIL.

$$\begin{aligned} (x + 3)(x + 4) &= x^2 + 4x + 3x + 12 \\ &= x^2 + 7x + 12 \end{aligned}$$

Step 2: This matches the original expression, so the factorization is correct.

Check: $x^2 + 7x + 12$ ✓ Matches!

8 QUICK REMINDERS & STRATEGY TIPS

GOOD STRATEGY:

- Always look for a GCF first.
- Check if it's a difference of squares.
- For trinomials $x^2 + bx + c$, find two numbers that multiply to c and add to b.
- For $ax^2 + bx + c$, use the AC method or grouping.
- After factoring, multiply the factors back to check your answer.

COMMON MISTAKES:

- Forgetting to factor out a GCF first.
- Using the wrong signs for b or c.
- Picking numbers with the right product but wrong sum.
- Forgetting both factors in a difference of squares.
- Not distributing correctly when multiplying to check.
- Forgetting to set each factor equal to zero when solving.

Back